Zhengshu Zhang

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EDUCATION

University of Southern California | Los Angeles, CA, USA **Jan 2025 – Dec 2026 (Expected)**

- M.S. Spatial Data Science | GPA: 3.88 / 4.00

Sichuan Agricultural University | Ya'an, Sichuan, CHN **Sep 2015 – Jun 2019**

- B.Eng. Food Quality and Safety | GPA: 3.14 / 5.00

PAPERS

Preprints & Manuscripts

- Zhang, Zhengshu and Li, Zhiyuan. "ARC-Bench: Closed-Loop Replanning Masks Broken Action Ranking in Frozen JEPA World Models." (August 12, 2026). arXiv: https://arxiv.org/abs/2609.05461 | arXiv:2609.05461 [cs.AI]. **ICLR 2027 under review.**
- Zhang, Zhengshu and Wang, Siqin. "Stop-Aware Reranking for Alternative Bus Routes: A GTFS-Grounded GeoAI Benchmark and LA28 Application."
- Zhang, Zhengshu, et al. "CINDER: Chronology-Aligned, Imbalance-Aware Neighborhood-Dynamic Encoding with Area-Budgeted Ranking for Annual Wildfire Forecasting." With Siqin Wang's group.
- Zhang, Zhengshu, Tang, Yimin, and Bıyık, Erdem. "RIFT: Repair of Infeasible Fixed-Endpoint Trajectories for Multi-Robot Kinodynamic Planning." In preparation for RA-L.
- Zhang, Zhengshu. "When Does Test-Time Physical Diagnosis Pay?" (September 14, 2026; v2 October 2026). arXiv: https://arxiv.org/abs/2609.22299 | arXiv:2609.22299 [cs.RO].

RESEARCH

World-Model Diagnosis: ARC-Bench | Group Research, Lead **Jun 2026 – Present**

In collaboration with Zhiyuan Li (Postdoctoral Researcher, Aalto University)

- Proposed and led ARC-Bench, a no-leak fixed-candidate protocol that measures whether frozen JEPA-style planning objectives rank candidate action sequences correctly; designed the audit across PointMaze, Wall, Push-T and Meta-World reach-wall, on officially released JEPA-WM checkpoints and on frozen DINOv2, V-JEPA 1 and V-JEPA 2 encoders.
- Measured the ranking apart from the planner: the top-scored candidate is suboptimal on **96.8%** of Push-T anchors and on **100%** of Meta-World reach-wall anchors, where Hit@1 is **0.0%**. Swapping DINOv2 for frozen V-JEPA encoders leaves the top pick wrong on **91.1–97.3%** of useful Push-T anchors, so the defect is not a property of one backbone.
- Showed that closed-loop replanning masks the defect: forcing longer open-loop blocks drops PointMaze success from **89.6%** to **61.8%** and Push-T from **36.1%** to **12.5%** over 144 paired episodes, with 44 PointMaze episodes rescued by frequent replanning against 4 lost (McNemar's p = 1.51e-9).
- Traced the fault to terminal-distance geometry rather than to any one checkpoint through topology-stress audits on public and procedural layouts (terminal-Euclidean ranking wrong on **100%** of U-Maze anchors, Spearman **−0.58**), under SHA-verified checkpoint provenance, undertraining, matched-budget and metric-circularity controls, with every paper-facing number traced to primary source files.

Multi-Robot Motion Planning: RIFT | Volunteer Researcher **Jul 2026 – Present**

Advisor: Prof. Erdem Bıyık | USC Viterbi School of Engineering

- Designed and implemented RIFT, a kinodynamic multi-robot motion planner that corrects all robots' controls together in iterations instead of replanning any robot; evaluated against db-CBS, db-ECBS, db-LaCAM, and K-ARC in their authors' implementations on random, swap, and cluttered-cross instances with unicycle and quadrotor models: RIFT solves every 64-robot swap instance where no baseline exceeds 50%, and runs at least 24× faster than db-ECBS on second-order robots.

- Conducted large-scale, preregistered falsification and repair campaigns involving hundreds of audited runs: identified root causes of solver failures in QP-based repair, line-search acceptance, and clearance representations, with every change versioned and independently audited.
- Investigated search- and learning-based MAPF methods (ITA-ECBS, RAILGUN, and Judgelight) to inform multi-arm robot coordination directions within the group.

Spin-Glass Optimization: Trees of Spin Clusters (TOSC) | Research Assistant **Jan 2026 – Aug 2026**

Advisor: Prof. Itay Hen | USC Information Sciences Institute – Hen Lab

- Developed and evaluated TOSC construction algorithms for Ising/spin-glass optimization, progressing from skeleton and frontier-absorption solvers to local-repair methods and repair-acceptance variants.
- Built a reproducible Python/NetworkX benchmarking pipeline across spin-tree and Chimera graph families, using all-start evaluation of coverage, objective cost, validity, and robustness against a baseline extracted from legacy C++ code.
- Designed a randomized, quality-gated Chimera Cell-Tree Cover method that improved targeted coverage–cost trade-offs while preserving quotient-tree validity.

GeoAI for Path Planning & Wildfire Forecasting | Research Assistant **Jan 2026 – Present**

Advisor: Prof. Siqin Wang | USC Spatial Sciences Institute

- **Bus-route benchmark** *(paper in final revision):* Designed a GTFS-grounded, multi-stage route-retrieval framework incorporating graph search, diversified K-shortest paths, stop-aware reranking, and learned reranking; improved Los Angeles Rank-1 F1 **from 0.260 to 0.438** and Best-of-3 F1 **from 0.345 to 0.457** over shortest-path baselines, with an LA–Portland cross-city transfer benchmark and LA28 venue-area case study.
- **Wildfire forecasting (CINDER)** *(in progress)*: Built a one-year-ahead burned-cell occurrence benchmark spanning 107,528 fixed 500-m cells across Los Angeles, San Diego, and Ventura Counties (2.37M cell-years; 38,288 positives), with chronology-aligned monthly predictors, spatially blocked validation (10-km folds; 5-km exclusion), and natural-prevalence out-of-time evaluation; led model design and experiments on masked temporal encoding, imbalance-aware training, and area-budgeted ranking.

ACADEMIC SERVICE

Course Grader: Spatial Data Science, under Prof. YI QI **Sep 2026 – Present**

SKILLS

- **Programming:** Python, C/C++, Java, SQL, JavaScript, TypeScript
- **Machine Learning:** PyTorch, latent world models, self-supervised learning, Transformers & attention, CNN/ResNet, RNN/LSTM, learning-to-rank (LambdaMART, CatBoost), reinforcement learning, spatial intelligence
- **Robotics & Planning:** MuJoCo, CEM/MPC planning, kinodynamic RRT, trajectory optimization, multi-agent path finding, spatial/GTFS data pipelines, multi-robot motion planning
- **Tools:** Git, Docker, Linux/WSL, AWS, GCP, FAISS/RAG, ArcGIS